

Prayas JEE 3.0 2025

Mathematics

DPP: 1

Inverse Trigonometric Functions

- Q1** If $x, y, z \in [-1, 1]$ such that $\cos^{-1} x + \cos^{-1} y + \cos^{-1} z = 3\pi$, then find the value of $xy + yz + zx$, is
 (A) 0 (B) 1
 (C) 3 (D) -3
- Q2** The range of $f(x) = [\tan^{-1} x]$, Where $[.]$ is greatest integer function.
 (A) $(-2, 2)$
 (B) $[-2, 2]$
 (C) $\{-2, -1, 0, 1, 2\}$
 (D) $\{-2, -1, 0, 1\}$
- Q3** Domain of $f(x) = \sqrt{\sin^{-1}(2x) + \frac{\pi}{6}}$ is:
 (A) $(-\frac{1}{2}, \frac{1}{2}]$
 (B) $[-\frac{1}{4}, \frac{3}{4}]$
 (C) $[-\frac{1}{4}, \frac{1}{4}]$
 (D) $[-\frac{1}{4}, \frac{1}{2}]$
- Q4** The range of $f(x) = \sin^{-1} x + \tan^{-1} x + \sec^{-1} x$ is
 (A) $(\frac{\pi}{4}, \frac{3\pi}{4})$
 (B) $[\frac{\pi}{4}, \frac{3\pi}{4}]$
 (C) $\{\frac{\pi}{4}, \frac{3\pi}{4}\}$
 (D) None of these
- Q5** $\cos^{-1}(\cos(\frac{5\pi}{4}))$ is given by:
 (A) $\frac{5\pi}{4}$
 (B) $\frac{3\pi}{4}$
- (C) $\frac{-\pi}{4}$
 (D) None of these
- Q6** The domain of $f(x) = \sin^{-1}(|x - 1| - 2)$ is:
 (A) $[-2, 0] \cup [2, 4]$
 (B) $(-2, 0) \cup (2, 4)$
 (C) $(2, 4)$
 (D) $[2, 4]$
- Q7** The domain of $\sin^{-1}[\log_3(\frac{x}{3})]$ is:
 (A) $[1, 9]$ (B) $[-1, 9]$
 (C) $[-9, 1]$ (D) $[-9, -1]$
- Q8** The value of $\cot(\operatorname{cosec}^{-1} \frac{5}{3} + \tan^{-1} \frac{2}{3})$ is
 (A) $\frac{6}{17}$
 (B) $\frac{3}{17}$
 (C) $\frac{4}{17}$
 (D) $\frac{5}{17}$
- Q9** The value of $\sin[\cot^{-1}\{\cos(\tan^{-1} 1)\}]$, is
 (A) $\frac{2}{3}$
 (B) $\sqrt{\frac{2}{3}}$
 (C) $\frac{1}{\sqrt{2}}$
 (D) $\sqrt{\frac{3}{2}}$
- Q10** If $\tan^{-1} \frac{x}{\pi} < \frac{\pi}{3}$, $x \in \mathbb{N}$, then the maximum value of x is
 (A) 2 (B) 5
 (C) 7 (D) None of these



Answer Key

Q1 (C)

Q2 (D)

Q3 (D)

Q4 (C)

Q5 (B)

Q6 (A)

Q7 (A)

Q8 (A)

Q9 (B)

Q10 (B)



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